AutoTerm-SST: Adaptive Terminology Memory with Zero Session-Time Setup for Streaming Speech Translation

Jiaxuan Luo[1] **Siqi Ouyang**[2] **Lei Li**[2*]
[1]Johns Hopkins University
[2]Carnegie Mellon University
jluo50@jhu.edu, {siqiouya,leili}@andrew.cmu.edu

Abstract

Retrieval-augmented speech translation improves rare terminology, but systems such as RASST still assume that users provide a glossary before each session. That assumption is inconvenient for live talks, while merging every domain into one growing glossary eventually places too many irrelevant terms in the same retrieval and prompt competition.

We present AutoTerm-SST, an interactive system that removes session-time glossary upload and domain selection. It organizes a large offline terminology catalog into topic slices, matches recent streaming translation context to bilingual slice prototypes, and activates a compact multi-slice working set under fixed retrieval and prompt budgets. On five ACL and five ESO oncology talks, AutoTerm matches a known-domain reference in terminology accuracy (88.3% vs. 88.1%) and mWER-resegmented BLEU (45.09 vs. 45.19), while requiring no domain input. Compared with a one-million-entry merged glossary, it improves terminology accuracy by 3.9 points and BLEU by 2.30. Routing plus four-slice retrieval remains lightweight: the complete system runs at 0.406 RTF, and the vLLM host sustains 32 real-time sessions. The live demo exposes active slices and retrieved terminology as the topic changes, providing zero-setup terminology-aware streaming translation.

1 Introduction

Simultaneous speech translation (SST) is increasingly used for technical talks, lectures, and live media. Modern speech-language models can translate long-form audio directly (Qwen Team, 2025; Ouyang et al., 2025), but rare methods, datasets, clinical concepts, and named entities remain difficult to infer from local acoustic context. Glossary retrieval addresses this problem by exposing relevant source–target term pairs to the language model before decoding.

RASST (Luo et al., 2026) shows that retrieval-augmented generation can improve out-of-distribution terminology in streaming translation. Its glossary, however, is fixed before a session. In practice, a listener may not know the talk domain in advance, and asking every user to find, upload, or select a glossary creates friction. A seemingly simple alternative is to merge all domains into a universal glossary. This works at moderate scale, but it is not a sustainable catalog policy: the retriever and prompt have fixed capacity, so irrelevant and conflicting mappings increasingly compete with useful evidence.

We introduce **AutoTerm-SST**, a system demo that moves glossary selection from user setup into the streaming loop. Operators register modular topic slices offline. During a session, a training-free router compares recent generated translation with bilingual slice prototypes and keeps several high-scoring slices inside a fixed term budget. Retrieval and language-model decoding remain unchanged; AutoTerm only controls which part of the catalog is allowed to compete for each prompt. Users neither upload a glossary nor select a domain.

This design provides a practical expansion path. New topic resources can be registered without retraining the translation model or forcing every session to search the entire catalog. The demo makes this behavior visible: users see the active working set, retrieved term pairs, and terminology-highlighted translation while audio is streaming.

Our contributions are:

- We present a zero-session-setup terminology-aware SST system built on RASST, with a live interface for inspecting active glossaries and retrieved evidence.
- We introduce a training-free, budgeted multi-slice memory that uses recent streaming translation context instead of user-provided domain metadata.

* Corresponding author.

- On a ten-talk ACL/ESO stream, we show the complete systems progression from retrieval-free InfiniSST, to known-domain and automatic working sets, to 100k and 1M universal glossaries; we also characterize retrieval/routing compute and verify 32-session real-time serving.

2 Related Work

Streaming speech translation. SST systems translate partial speech while balancing quality and latency. Prefix-to-prefix policies make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), and SimulEval standardizes streaming evaluation (Ma et al., 2020). Recent speech-language models broaden the modeling choices for direct SST (Seamless Communication, 2023; Zhang et al., 2024; Qwen Team, 2025). AutoTerm-SST does not propose a new translation model; it contributes a terminology-management and demo layer around an existing streaming model.

Terminology-aware translation. Terminology control is commonly implemented with user-provided dictionaries, constrained decoding, or training-time exposure to constraints (Hokamp and Liu, 2017; Dinu et al., 2019). Terms and named entities remain a known weakness of speech translation (Gaido et al., 2021). Retrieval-augmented generation offers a non-parametric alternative (Lewis et al., 2020; Khandelwal et al., 2021), and RASST applies glossary retrieval before streaming LLM decoding (Luo et al., 2026). These approaches normally assume that the relevant glossary is known before translation. AutoTerm instead manages a budgeted cache of topic slices online.

Interactive translation systems. Serving engines such as vLLM (Kwon et al., 2023) and SGLang (Zheng et al., 2024) provide continuous batching, while interactive translation tools make model behavior inspectable (Dandekar et al., 2024). AutoTerm-SST combines streaming audio transport with a live terminology evidence view and automatic glossary selection.

3 System Overview

AutoTerm-SST wraps the RASST speech encoder and MaxSim term retriever (Luo et al., 2026), together with the Qwen3-Omni language model with an adaptive terminology loop (Figure 1). For chunk t, the speech encoder produces h_t and the retriever[1] searches only the current working set. At most ten source–target term pairs are placed in the language-model prompt. After decoding, the new translation is appended to a short context window and may change the working set for a later chunk.

The adaptive step is outside the translation model. It neither changes the speech representation nor grants the retriever a larger candidate budget. Instead, it decides which prebuilt indexes share that budget. This separation keeps the system pluggable: the same streaming protocol can serve a retrieval-free agent, a fixed-glossary RASST agent, or AutoTerm, while the web interface receives structured evidence with every partial translation.

AutoTerm enforces three invariants: the live request contains no domain or glossary; catalog growth cannot increase per-chunk retrieval or prompt budgets; and a new topic is an offline data update, not model training. All policies therefore share the speech model, retriever, decoder, and streaming schedule, and differ only in which terminology indexes are eligible.

The live agent emits the translated text together with retrieved term pairs, their source slices, retrieval latency, and the current working-set state. These fields drive the evidence panel in Section 5 and make terminology behavior observable rather than hidden inside a prompt.

4 Adaptive Terminology Memory

AutoTerm separates the broad offline catalog from the compact glossary exposed to a live session. Each topic slice contains term–translation pairs, a prebuilt MaxSim index, and a bilingual prototype formed from a short topic description and representative terminology. Domain-dependent mappings are retained: the same source term may have different translations in different slices.

Offline catalog registration. An operator registers each slice through a manifest containing its label, term count, prototype, and prebuilt index. Slices are independently replaceable; topic-dependent mappings remain separate. At session time the router reads the compact manifest and prototypes, while indexes outside the working set do not participate in retrieval. Catalog capacity is

[1] RASST pre-encodes each source term as a normalized BGE-M3 vector z_i. For speech-window vectors q_w, it scores $r_i = \max_{w \in \mathcal{W}_t} q_w^\top z_i$, then attaches the stored target translation.

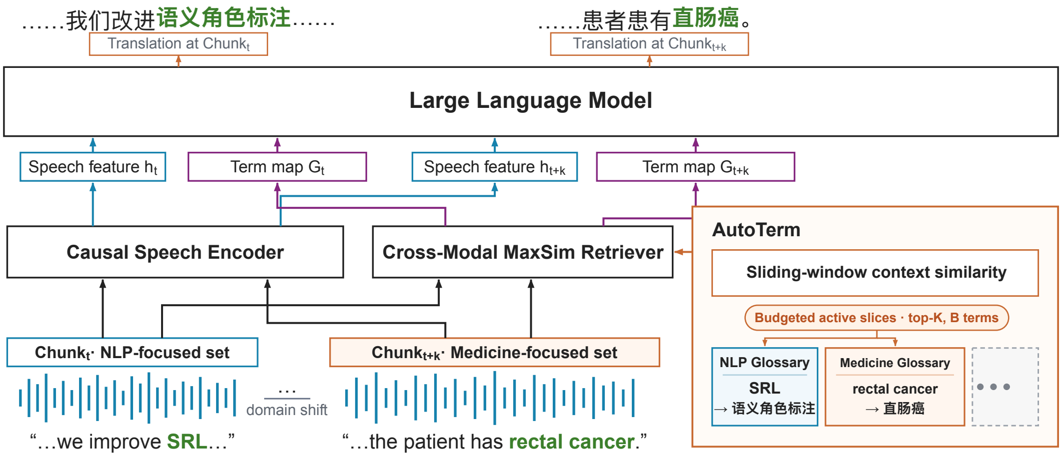

Figure 1: AutoTerm-SST as a stream moves from NLP to medicine. Recent translation context selects several glossary slices under a fixed term budget; RASST retrieves prompt evidence only from this working set. The ellipsis denotes additional registered topic slices.

thereby separated from the terminology exposed to one decoding step.

Semantic routing. At chunk t, the router concatenates the most recent w generated translation chunks into context c_t. For slice d with prototype examples P_d and multilingual sentence encoder E, it computes

$$\bar{p}_d = \mathrm{norm}\left(\frac{1}{|P_d|}\sum_{p\in P_d} E(p)\right), \qquad (1)$$
$$s(d \mid c_t) = \cos(E(c_t), \bar{p}_d)\,.$$

We use BGE-M3 (Chen et al., 2024) for E. Prototypes contain a short bilingual description and representative terms, so Chinese generated context and English topic metadata share one embedding space. The routing path is therefore a direct semantic similarity computation over accumulated translation context, not a classifier trained on the evaluation talks.

Why generated target context? The speech-language model translates audio directly, without a stable source transcript. Its partial target text is already available and becomes more topic-specific over time, enabling comparison with bilingual prototypes without a supervised domain model. Routing is causal: chunk t only changes later retrieval. A generic early translation can therefore delay topic evidence; accumulating four chunks and retaining several slices makes this signal less brittle.

The highest-scoring slices are admitted until either the maximum slice count K or total active-term budget B is reached. The current best slice is pinned when possible to reduce oscillation. Keeping multiple plausible slices is important: a temporary top-1 routing error need not remove the correct terminology from retrieval.

Working-set update. At each routing tick, AutoTerm keeps the current slice if it fits, then fills remaining slots in similarity order. Lower-ranked non-pinned slices stay outside the budget. This cache-like rule is not a hard domain switch: plausible topics can coexist during a transition, while stale slices disappear as their scores fall. Candidate indexes are preloaded and selections change only between decoding steps.

Fixed evidence budget. All active slices share a top-100 returned-candidate budget and a common top-10 prompt cap. Increasing K therefore changes the composition of the search space, not the amount of evidence available to AutoTerm. In the evaluated configuration, the router considers ten 1k slices, uses the latest four translation chunks, and activates at most four slices (B=4,000). The known-domain reference searches the matching 1k slice; universal baselines search nested 100k and 1M indexes under exactly the same retrieval and prompt limits.

For a multi-slice query, the system divides the 100 candidate slots among the active indexes, combines their results, removes only exact duplicate source–target pairs, and globally reranks the remaining candidates before the top-10 prompt cap. Consequently, selecting four slices does not give AutoTerm four times the retrieval or prompt ca-

Setting	Input	Active	TERM_ACC (%)			BLEU	MT-BLEU	Prec. (%)	Refs
			NLP	Med.	All				
InfiniSST	None	0	76.6	46.3	61.4	41.84	38.32	0.0	0.00
Known-domain-1k	Domain	1k	88.8	**86.6**	**88.1**	**45.19**	**42.39**	**48.7**	**0.69**
AutoTerm-1k×4	None	≤4k	89.3	86.3	**88.3**	45.09	42.28	34.8	0.94
Merged-100k	None	100k	**89.4**	84.5	87.8	43.72	40.93	6.8	4.96
Merged-1M	None	1M	87.1	78.9	84.4	42.79	39.95	3.4	8.70

Table 1: Ten-talk comparison under a fixed retrieval/prompt budget. BLEU and MT-BLEU use StreamLAAL-compatible mWER resegmentation. The four RAG rows share 2,047 raw-term occurrences (1,368 NLP; 679 medicine). InfiniSST reports the canonical LM=2 five-talk scores; its All/BLEU/MT-BLEU are two-corpus macro means. Input is session-time user setup (Domain means manual selection; None means zero setup); Active is the maximum glossary entries searched per chunk. Prec. is prompt precision; InfiniSST denotes retrieval-free RASST (Ouyang et al., 2025) and injects no references.

pacity. Its advantage can only come from placing more relevant mappings inside the same evidence budget.

The context encoder scores all lightweight prototypes together, and likely indexes are warmed asynchronously. Each chunk retrieves from a snapshot of the working set installed before that chunk began; a routing update that finishes during generation only affects the next chunk. This keeps terminology evidence causal and prevents cold index loading from changing an in-flight prompt.

The million-entry baseline is a capacity stress condition, not a claim that large-index search is computationally impossible. AutoTerm addresses a different problem: preserving relevant prompt evidence as an offline catalog grows, without requiring any session-time input from the user.

5 Demo Interface

The demo begins like an ordinary streaming translator: the user selects a language pair and starts a microphone or file stream. In AutoTerm mode there is no glossary upload or domain selector. The system starts with a small working set and updates it as translation context accumulates.

Walkthrough. The page first shows live partials while context accumulates. The working-set badge then updates to the most similar slices, and retrieved pairs appear beside the partial that used them. After a topic change, old and new slices can briefly coexist until the sliding context reflects the new subject. The user changes no control throughout this progression.

Visible terminology evidence. The interface highlights retrieved terminology in the translation and lists the corresponding source–target mappings. A compact status row shows the active slices, term count, and retrieval latency. Thus a viewer can tell whether a mistranslation came from missing retrieval evidence, an irrelevant slice, or language-model decoding.

Extensibility. Operators register or replace topic slices offline; users see the benefit in the next session without model training or UI changes. Fixed-glossary presets and ad-hoc term overrides remain available for expert users, but they are not needed for the zero-setup workflow.

Registration validates term pairs, builds an independent index, and adds one manifest entry. Existing sessions retain their current snapshot, while new sessions immediately see the updated candidate set. Thus an organization can grow or revise its catalog without restarting the translation service or rewriting the client application.

Availability. The MIT-licensed implementation, reviewer-friendly launcher, and screencast are available in the GitHub repository. The hosted entry page is `https://luojiaxuan.github.io/autoterm-sst/`; live mode uses Qwen3-Omni with vLLM continuous batching and the RASST retriever.

6 Evaluation

Setup. We alternate five English ACL 60/60 dev talks with human-annotated terminology (Salesky et al., 2023) and five long-form English oncology videos from the European School of Oncology (ESO) test set (Iranzo-Sánchez et al., 2025) in one 4.7-hour En→Zh stream. Following RASST (Luo et al., 2026), we reuse its GPT-5.4-generated Chinese ESO references and GPT-extracted, manually verified terminology. All conditions use RASST's

Figure 2: The live En→Zh interface after an automatic transition to medicine. Retrieved term pairs, source slices, and translation highlights expose the evidence used for each partial output.

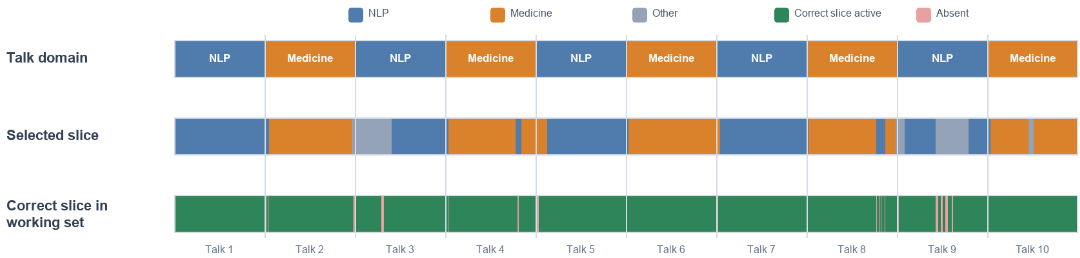

Figure 3: AutoTerm routing across ten alternating talks. The selected slice may briefly lag a topic change, but the correct glossary remains available inside the four-slice working set for 96.9% of decoder windows. Talk widths are normalized for readability.

Qwen3-Omni streaming formulation and 1.92 s chunks. *InfiniSST* (Ouyang et al., 2025) uses a Qwen3-Omni speech model after supervised fine-tuning for multi-turn, chunkwise streaming translation; MaxSim and term-map injection are disabled. The four RAG conditions share the retriever, top-100 returned-candidate budget, threshold 0.78, and top-10 prompt cap. We compare it with *Known-domain-1k*, which requires the talk domain; *AutoTerm-1k×4*, which requires no session input; and *Merged-100k/1M*, two nested universal glossaries.

The ten registered AutoTerm slices contain 1k entries each. The merged baselines combine the same target-bearing resources with distractors to exactly 100k and 1M mappings; 100k is a strict prefix of 1M. Exact duplicate pairs are removed, but conflicting domain-dependent translations are retained. Router settings were selected on four windows drawn from these talks before the complete run. Because the registered NLP and medicine slices include benchmark terminology, this is a controlled routing and retrieval-competition evaluation, not a held-out glossary-induction test.

Comparison provenance. The InfiniSST row reuses archived outputs for these exact ten talks and the matching 1.92 s policy. It is the system retrieval ablation, not a separate legacy architecture. We retain its canonical five-talk ACL and ESO scores and macro-average the two corpus reports. The other four rows come from one audited run per policy with identical audio order and chunk boundaries. Corpus BLEU is computed after pooling the resegmented sentences, rather than by averaging the ACL and medicine BLEU scores.

Metrics. Following RASST (Luo et al., 2026), **TERM_ACC** is the exact-match rate of target translations in aligned hypotheses over a fixed raw-term denominator: 2,047 time-aligned occurrences (1,368 NLP; 679 medicine) for every RAG policy. Source aliases sharing target variants count once; nested terms and distinct translations remain separate. InfiniSST retains its canonical raw-term scorers. **BLEU** is corpus SacreBLEU (Papineni et al., 2002; Post, 2018) after mWERSegmenter (Matusov et al., 2005) resegments each talk to StreamLAAL-compatible reference boundaries (Papi et al., 2024). **Masked-term BLEU**

(MT-BLEU) repeats this scoring after masking the full corpus raw target inventory in both hypothesis and reference, not the UI highlight subset. Prompt precision is the fraction of injected references that overlap a local raw-term occurrence; Refs is the mean number of references injected per chunk. The masking inventory is fixed across policies.

From InfiniSST to an adaptive working set. AutoTerm exceeds InfiniSST by 26.9 TERM_ACC points, 3.25 BLEU, and 3.96 MT-BLEU, yet remains within 0.24 TERM_ACC and 0.11 BLEU/MT-BLEU of the known-domain reference. Merged-100k remains usable, but AutoTerm gains 1.37 BLEU and has 34.8% rather than 6.8% prompt precision. At 1M, its advantage grows to 3.91 TERM_ACC and 2.30 BLEU points.

Where does the scale loss occur? Merged-100k remains competitive on NLP and trails AutoTerm by only 1.8 points on medicine. At 1M, the medicine gap grows to 7.4 points and an NLP gap also appears. Thus Table 1 does not claim every large index fails; under fixed retrieval and prompt capacity, scale increases sensitivity to domain distribution and ambiguity, which a compact routed set avoids.

Why does translation quality move? AutoTerm injects 0.94 references per chunk at 34.8% precision, versus 4.96 at 6.8% for Merged-100k and 8.70 at 3.4% for Merged-1M. Larger indexes therefore fill a fixed prompt with more low-value mappings. AutoTerm still leads 1M by 2.33 MT-BLEU points after masking evaluated term translations, showing that noise also affects surrounding translation.

Routing behavior. Figure 3 illustrates why AutoTerm uses a multi-slice working set rather than a hard top-1 router. Transient top-slice errors occur around topic boundaries, yet another active slice usually preserves the correct terminology inventory. This limits the cost of delayed topic evidence in the generated translation context.

Observed failure modes. Two audited failures clarify the limits. After an ACL-to-medicine boundary, stale context temporarily kept the medicine slice outside the working set. In a medicine segment about research papers, Merged-1M supplied mappings for both a scholarly article and physical paper, whereas AutoTerm retained the contextual sense. Routing therefore reduces, but does not eliminate, delayed or ambiguous evidence.

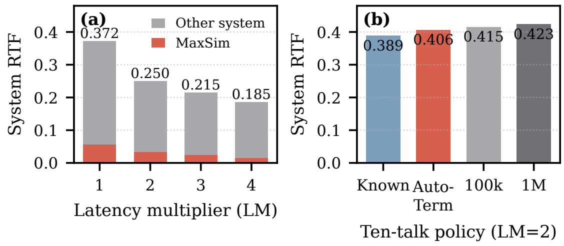

Figure 4: Standard system RTF (wall time/audio duration; real time is below 1). (a) RASST's LM sweep, with MaxSim inside the total. (b) Warm 4.7-hour ten-talk runs on one H200 at LM=2.

System and serving scale. System RTF is processing wall time divided by input audio. The RASST LM 1–4 sweep yields 0.372, 0.250, 0.215, and 0.185; MaxSim is respectively 15.0%, 13.4%, 11.1%, and 8.2% of total wall time (Figure 4(a)). On the current single-H200 4.7-hour stream at LM=2, known-domain, AutoTerm, Merged-100k, and Merged-1M obtain 0.389, 0.406, 0.415, and 0.423 (Figure 4(b)). These warm measurements span WebSocket connection through the final cursor and include routing, retrieval, generation, scheduling, and transport. AutoTerm is 2.46× real time; its 10.6% routing/retrieval share is a component ratio, not another system RTF. On 2×A6000, vLLM generation p95 remains 0.81–0.89 s for 8–32 real-time clients.

7 Conclusion

AutoTerm-SST provides zero-setup terminology retrieval over a modular offline catalog. On mixed ACL and ESO oncology talks, it matches known-domain quality, beats InfiniSST, avoids 1M universal-glossary noise, and makes this behavior visible in the demo.

Limitations

The En→Zh slices contain benchmark terms, so we measure routing and retrieval competition rather than glossary induction. Routing can lag topic changes, and scale sensitivity depends on language and budgets; 100k and 1M are not universal thresholds. We evaluate AutoTerm only on En→Zh. Although the underlying RASST translator was evaluated on En→Zh/De/Ja (Luo et al., 2026), AutoTerm routing remains unvalidated for other target languages and non-English source speech.

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